Probability Basics

CSU, Chico Math 350

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Definitions

Here are some important terms to know for Probability (from Ch 2.1, Definition 2.1):

- An **experiment** is a process that produces an **observation**.
- An **outcome** is a possible observation.
- The set of all possible outcomes is called the **sample space**, commonly denoted $\mathcal{S}$
- An **event** is a subset of the sample space.
- A trial is a single running of an experiment.

Sample Space, examples

Here are some examples of sample spaces:

- two hyperlinks that are classified as working (W) or broken (B)
  - $\{WW, WB, BW, BB\}$
- survey answer for the amount of time (in months, say) until a new textbook edition releases.
  - $\{0, 1, 2, \ldots\}$

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Set Operations

Here's a few official definitions for some set operations:

union

The union of k sets A_j is the event consisting of A_1 or (not xor) A_2 or $\ldots$ A_k.
Mathematically, we write $\cup_{j=1}^{k} A_j$.
This is also thought of as **at least one of A_j happens.**

intersection

The intersection of k sets A_j is the event consisting of A_1 and A_2 and $\ldots$ A_k.
Mathematically, we write $\cap_{j=1}^{k} A_j$.
This is also thought of as **all of $A_1, \ldots, A_k$ happen.**

Set Operations

complement

The complement A^c of A is the set of events that **are not in** A**.**

difference

The difference $B - A$ is the set of events **contained in** B **but not in** A**.**

Set Operations

disjoint

Two sets A and B are disjoint (or mutually exclusive) if A and B do not overlap, i.e. **have no events in common.**
Mathematically, we write $A \cap B = \emptyset$.

subset

The set A is said to be a subset of B if **all events in A imply an event of the set B.**
Mathematically, we write $A \subseteq B$.
We also write $a \in A$ **implies** $a \in B$.

Subsets

Some notes on subsets:

- Sometimes people will use the symbol $\subset$ to explicitly state that A is a subset of B, but certainly not equal to B.
- To show that $A = B$, one needs to show both that $x \in A$ implies $x \in B$ and that $x \in B$ implies $x \in A$.

Laws of Set Operations

There are well established laws of sets, analogous to laws of mathematics.

- Commutative: $A \cup B = B \cup A$ and
  $A \cap B = B \cap A$.
- Associative: $(A \cup B) \cup C = A \cup (B \cup C)$ and
  $(A \cap B) \cap C = A \cap (B \cap C)$.
- Distributive: $C \cap (A \cup B) = (C \cap A) \cup (C \cap B)$ and
  $C \cup (A \cap B) = (C \cup A) \cap (C \cup B)$.
- De Morgan's Laws: $(A \cup B)^c = A^c \cap B^c$ and
  $(A \cap B)^c = A^c \cup B^c$.

Equally Likely Outcomes

The most common probability models are those consisting of finite sample spaces, where each of N outcomes is equally likely to occur:

- coin flip: $\mathcal{S} = \{H, T\}$, $P(H) = 1/2$
- six-sided die roll: $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$, $P(\text{roll } 1) = 1/6$
- If N equally likely outcomes are in $\mathcal{S}$, then the probability of each outcome **O** is $P(O) = 1/N$

Probability, example

Imagine you roll two (fair) dice. What is the probability of the event $E = \{R_1 + R_2 = 7\}$?

- What is the sample space? Note the order of your rolls!
  - $\mathcal{S} = \{(1,1),(1,2),\dots,(6,6)\}$ implies $N = 36$.
- How many ways can we roll a sum of 7?
  - $E = \{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\}$, so we see that six outcomes satisfy this event.
- The elements of E are disjoint and equally likely

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Probability Axioms

For a given sample space $\mathcal{S}$:

- Probabilities are **non-negative real numbers** $P(E) \geq 0$ for all events E.
- The probability of the sample space is 1. $P(\mathcal{S}) = 1$.
- Probabilities are countably additive. For pairwise disjoint events $A_1, A_2, ...,$ $\cup_{j=1}^{\infty} A_j = \sum_{j=1}^{\infty} P(A_j)$

Additional Properties

- $P(\emptyset) = 0$
- If A and B are disjoint, then $P(A) + P(B) = P(A \cup B)$
- If $A \subset B$, then $P(A) \leq P(B)$
- $0 \leq P(A) \leq 1$
- $P(A) = 1 - P(A^c)$
- $P(A - B) = P(A) - P(A \cap B)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

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Frequentist vs. Evidential Characterization

- Under the **frequentist** framework, suppose that we could conduct the same experiment over and over again, each time recording whether some event of interest took place. The proportion of occasions on which the event took place would converge to some number between 0 and 1, which would then be defined as the probability of the event.
- A classic example is flipping a coin repeatedly. The probability of the coin coming up heads is defined as $1/2$ if, in the long run, the proportion of flips yielding heads approaches $1/2$.
- However, the **evidential** framework instead measures the degree of certainty about whether an event occurs.
- Applied to the coin flip case above, if you flip a coin but don't observe the result and aren't repeating the random process of additional flips, your degree of certainty about the outcome being heads remains $1/2$.

Odds vs. Probabilities

- These are two different things! The odds of an event E are $\frac{P(E)}{1-P(E)}$
- For a six sided die, the probability of the event E of rolling a 1 is $P(E) = 1/6$.
- But the **odds** of E would be $\frac{P(E)}{1-P(E)} = \frac{(1/6)}{(5/6)}$.
- This would be written "1:5" or one-to-five odds of rolling a 1 versus rolling anything else.

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references I

Speegle & Clair. Probability, Statistics, and Data: A Fresh Approach Using R, 2021.

Also adapted from notes by Dr. Roualdes.